00 - Text - Preprocessing - Pitt

In this notebook, we will pre-process text files in Pitt set and split them into train / dev / test set with text.

In [1]:
```
import pandas as pd
```

In [2]:
```
fname_train = "../data/pitt/pitt_train.csv"
fname_dev = "../data/pitt/pitt_dev.csv"
fname_test = "../data/pitt/pitt_test.csv"
fname_pitt_text = "../data/pitt/pitt.csv"
```

In [3]:
```
train_df = pd.read_csv(fname_train)
dev_df = pd.read_csv(fname_dev)
test_df = pd.read_csv(fname_test)
text_df = pd.read_csv(fname_pitt_text)
```

In [4]:
```
# quick look at the dataframes
print("Train len =", len(train_df))
train_df['id'] = train_df['id'].astype(str)
train_df.head()
```

```
Train len = 174
```

Out[4]:

	id	target	age_of_first_test	sex	less_than_65
0	714	ad	79	male	False
1	466	ad	78	female	False
2	511	ad	75	female	False
3	609	ad	70	female	False
4	504	ad	78	male	False

In [5]:
```
print("Dev len =", len(dev_df))
dev_df['id'] = dev_df['id'].astype(str)
dev_df.head()
```

```
Dev len = 59
```

Out[5]:

	id	target	age_of_first_test	sex	less_than_65
0	234	ad	66	male	False
1	256	cn	67	male	False
2	497	ad	70	female	False
3	264	ad	65	female	False
4	297	cn	64	male	True

In [6]:
```
print("Test len =", len(test_df))
test_df['id'] = test_df['id'].astype(str)
test_df.head()
```

```
Test len = 59
```

Out[6]:

	id	target	age_of_first_test	sex	less_than_65
0	358	ad	60	female	True
1	34	cn	70	male	False
2	42	cn	56	female	True
3	650	ad	87	female	False
4	332	cn	50	male	True

In [7]:
```
print("Pitt text len =", len(text_df))
text_df.drop(text_df.columns[0], axis=1, inplace=True)
text_df.head()
```

```
Pitt text len = 549
```

Out[7]:

	id	file	target	text
0	1	001-0.cha	AD	mhm .\nalright .\nthere's &-um a young boy tha...
1	1	001-2.cha	AD	mhm .\nthere's a young boy &-uh going in a coo...
2	3	003-0.cha	AD	here's a cookie jar .\nand the lid is off the ...
3	5	005-0.cha	AD	the boy is slipping off the stool .\nhe's tryi...
4	5	005-2.cha	AD	okay ‡ he's fallin(g) off a chair .\nshe's &-...

In [8]:
```
def get_split_name(id):
    """
        return split name (train/dev/set) given id
    """
    result = "train" # default
    id = str(id)
    if len(dev_df.query("id == @id")) > 0:
        result = "dev"
    elif len(test_df.query("id == @id")) > 0:
        result = "test"

    return result

assert get_split_name("714") == "train"
assert get_split_name("466") == "train"
assert get_split_name("511") == "train"

assert get_split_name("234") == "dev"
assert get_split_name("256") == "dev"
assert get_split_name("497") == "dev"
```

```
assert get_split_name("358") == "test"
assert get_split_name("34") == "test"
assert get_split_name("332") == "test"

print("Passed")
```

```
Passed
```

In [9]:
```
# apply to text df
text_df["split_name"] = text_df["id"].apply(lambda id: get_split_name(id))
text_df.head()
```

Out[9]:

	id	file	target	text	split_name
0	1	001-0.cha	AD	mhm .\nalright .\nthere's &-um a young boy tha...	test
1	1	001-2.cha	AD	mhm .\nthere's a young boy &-uh going in a coo...	test
2	3	003-0.cha	AD	here's a cookie jar .\nand the lid is off the ...	train
3	5	005-0.cha	AD	the boy is slipping off the stool .\nhe's tryi...	train
4	5	005-2.cha	AD	okay ǂ he's fallin(g) off a chair .\nshe's &-...	train

In [10]:
```
assert text_df.query("id == 609")["split_name"].iloc[0] == "train"
assert text_df.query("id == 504")["split_name"].iloc[0] == "train"

assert text_df.query("id == 264")["split_name"].iloc[0] == "dev"
assert text_df.query("id == 297")["split_name"].iloc[0] == "dev"

assert text_df.query("id == 650")["split_name"].iloc[0] == "test"
assert text_df.query("id == 332")["split_name"].iloc[0] == "test"
print("Passed")
```

```
Passed
```

In [11]:
```
# split to 3 files
train_fname_out = "../data/pitt/pitt_train_txt.csv"
dev_fname_out = "../data/pitt/pitt_dev_txt.csv"
test_fname_out = "../data/pitt/pitt_test_txt.csv"

txt_train_df = text_df.query("split_name == 'train'")
txt_dev_df = text_df.query("split_name == 'dev'")
txt_test_df = text_df.query("split_name == 'test'")
```

In [12]:
```
txt_train_df.head()
```

Out[12]:

	id	file	target	text	split_name
2	3	003-0.cha	AD	here's a cookie jar .\nand the lid is off the ...	train
3	5	005-0.cha	AD	the boy is slipping off the stool .\nhe's tryi...	train
4	5	005-2.cha	AD	okay ‡ he's fallin(g) off a chair .\nshe's &-...	train
5	7	007-1.cha	AD	xxx .\ncan I look at it and tell you ?\noh oka...	train
6	7	007-3.cha	AD	well the girl is telling the boy to get the co...	train

In [13]:
```
txt_dev_df.head()
```

Out[13]:

	id	file	target	text	split_name
13	16	016-0.cha	AD	guy or a boy reaching for a cookie jar &-uh ti...	dev
14	16	016-1.cha	AD	uh the water's spilling over .\ndrying the dis...	dev
15	16	016-3.cha	AD	a boy is reaching for the cookie jar and tripp...	dev
16	16	016-4.cha	AD	well the little girl wants the cookie that the...	dev
26	33	033-0.cha	AD	okay .\napparently a mother and a daughter and...	dev

In [14]:
```
txt_test_df.head()
```

Out[14]:

	id	file	target	text	split_name
0	1	001-0.cha	AD	mhm .\nalright .\nthere's &-um a young boy tha...	test
1	1	001-2.cha	AD	mhm .\nthere's a young boy &-uh going in a coo...	test
18	23	023-0.cha	AD	(..) well I see the mother .\nwhat's she doin(...	test
19	23	023-2.cha	AD	well you see <the little> the little boy stea...	test
33	39	039-0.cha	AD	okay we see &-um first of all &-uh the sink is...	test

In [15]:
```
# write to csv files
print("Writing to csv files...")
txt_train_df.to_csv(train_fname_out,
                    index=False
                   )
txt_dev_df.to_csv(dev_fname_out,
                  index=False
                 )
txt_test_df.to_csv(test_fname_out,
                   index=False
                   )

print("Completed")
```

```
Writing to csv files...
Completed
```